"UG" END SEMESTER EXAMINATION – FEBRUARY, 2023

(REGULAR – ODD SEMESTER)

1ST YEAR (1ST SEMESTER) – MECHANICAL ENGINEERING

ENGINEERING MECHANICS

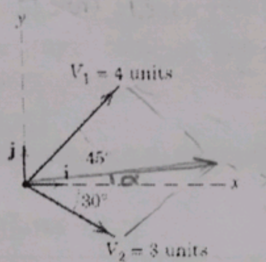
MENUGES01

[Full Marks : 80] [Time : 3:00 hrs.]

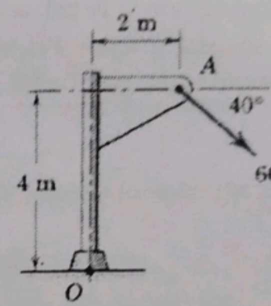
Assume suitable data(s), if not provided

A. Fill in the blanks/Multiple choice questions (*Compulsory*) – $[10X2 = 20]$

1. The unit vectors along the direction of resultant vector (**Fig – A.1**) are – .... & ....



R=4.33.



600cos40

Figure – A.1 **Figure – A.2**

2. The moment about the base point O of the 600 N force (**Fig. – A.2**) in vector form is – ....

3. The point, through which the whole weight of the body acts, is known as – .... Centre of Gravity.

4. If the resultant of two forces P and Q acting at an angle θ makes an angle α with P, then the resultant and the angle of resultant should be – .... & ....α......., respectively.

5. If two forces each equal to P in magnitude act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is – ....

6. A framed structure is perfect if it contains members equal to – (n = *number of joints*)
   a) $2n-3$ b) $3n-2$ c) $2n-1$ d) $n-2$

7. In the **Fig. – A.3**, angle α compared to β will be –
   a) Greater b) Lower
   c) Same d) Unpredictable

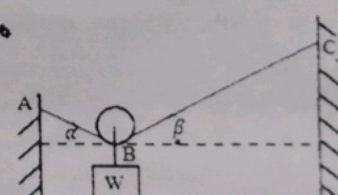


Figure – A.3

8. Force $5i + 6j + 7k$ N acting with a position vector is $-8i - 9k$ N. The resultant moment will be of – ....

9. The three forces keep acting at a point are in equilibrium, then –
   a) The forces must be coplanar b) The forces need not be coplanar
   c) The forces must be of equal magnitude d) The forces must be of unequal magnitude.

10. A mass of 10 kg is resting on a rough table. The friction force acting on it is – ....D..............

B. Short questions (*answer any four questions*): [$5X4 = 20$]

1. Two tugboats are pulling a barge (**Figure – B.1**). If the resultant of the forces exerted by the tugboats is a 5000-kg force directed along the axis of the barge, determine –
   a) The tension in each of the ropes, given that $\alpha = 45^0$.
   b) The value of α for which the tension in rope 2 is minimum.

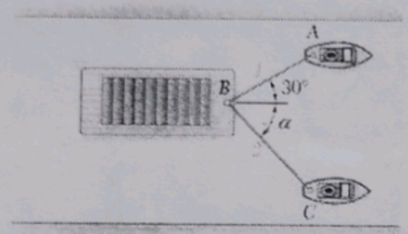


Figure – B.1

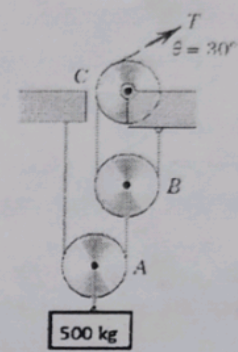


Figure – B.2

2. Calculate the tension T in the cable which supports the 500-kg mass with the pulley arrangement shown in **Figure – B.2**. Each pulley is free to rotate about its bearing, and the weights of all parts are small compared with the load. Find the magnitude of the total force on the bearing of pulley C.

3. Locate the centroid of the shaded area. (**Figure – B.3**) (all are in mm.)

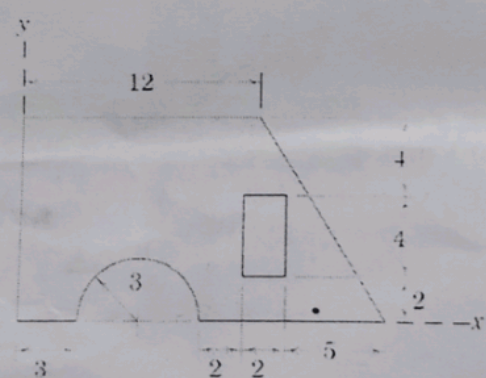


Figure – B.3

4. The position coordinate of a particle which is confined to move along a straight line is given by $s = 2t^3 - 24t + 6$, where s is measured in meters from a convenient origin and t is in seconds. Determine –
   a) The time required for the particle to reach a velocity of $72\ m/s$ from its initial condition at $t = 0$,
   b) The acceleration of the particle when $v = 30\ m/s$,

5. The $3 - kg$ slider is released from rest at position 1 (**Figure – B.4**) and slides with negligible friction in a vertical plane along the circular rod. The attached spring has a stiffness 350 N/m and has an un-stretched length of 0.6 m. Determine the velocity of the slider as it passes position 2.

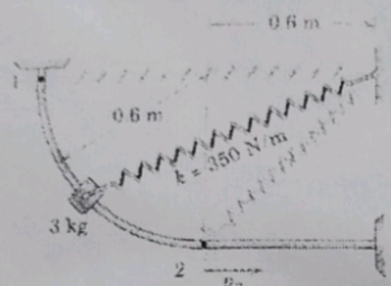


Figure – B.4

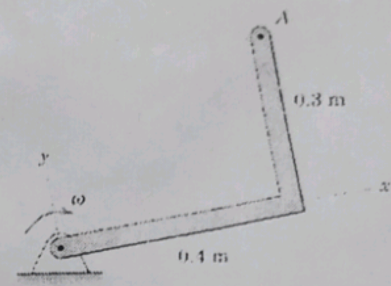


Figure – B.5

5. The right-angle bar (**Figure – B.5**) rotates clockwise with an angular velocity which is decreasing at the rate $4\,rad/sec^2$. Write the vector expressions for the velocity and acceleration of point A when $\omega = 2\,rad/sec$.

Long questions (*answer any four questions*): $[10X4 = 40]$

1. Determine the resultant of the four forces and one couple which act on the plate shown in **Figure C.1**.

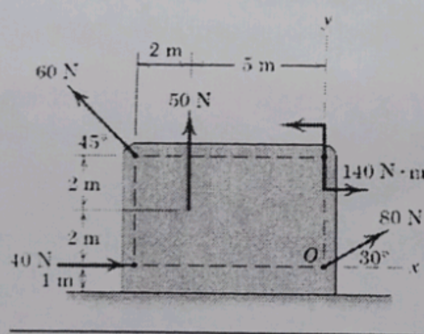


Figure – C.1

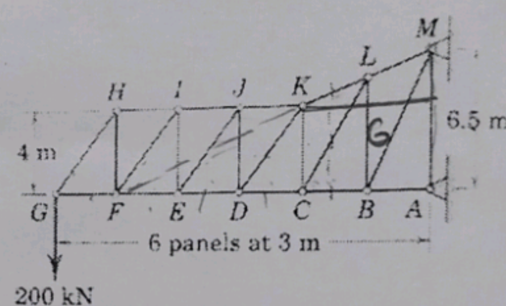
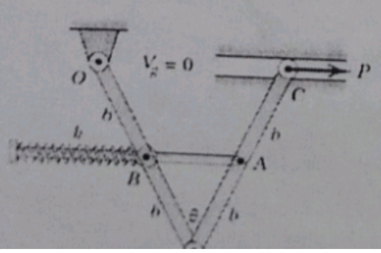
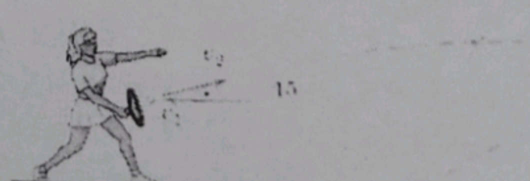


Figure – C.2

2. Calculate the forces induced in members KL, CL, and CB by the 200-kN load on the cantilever truss. (**Figure – C.2**)

3. The two uniform links, each of mass m, are in the vertical plane and are connected and constrained as shown in **Figure C.3**. As the angle θ between the links increases with the application of the horizontal force P, the light rod, which is connected at A and passes through a pivoted collar at B, compresses the spring of stiffness k. If the spring is uncompressed in the position where $\theta = 0$, determine the force P which will produce equilibrium at the angle θ.





4. A tennis player strikes the tennis ball with her racket when the ball is at the uppermost point of its trajectory as shown in **Figure C.4**. The horizontal velocity of the ball just before impact with the racket is $v_1 = 15\ m/s$ and just after impact its velocity is $v_2 = 21\ m/s$ directed at the 15^0 angle as shown. If the $60\ gm$ ball is in contact with the racket for $0.02\ s$, determine the magnitude of the average force R exerted by the racket on the ball. Also determine the angle β made by R with the horizontal.

5. Automobile A is travelling east at the constant speed of $36\ km/h$. As automobile A crosses the intersection shown in **Figure C.5**, automobile B starts from rest 35-m north of the intersection and moves south with a constant acceleration of $1.2\ m/s^2$. Determine the position, velocity, and acceleration of B relative to A 5-sec after the intersection. (**Figure – C.5**)

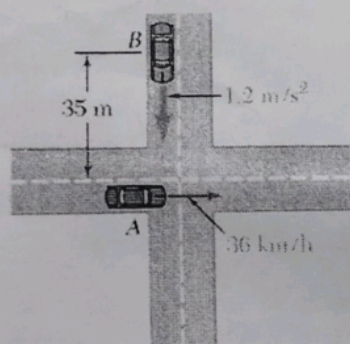


Figure – C.5

6. The angular acceleration of a flywheel is given by $\alpha = 12 - t$, where, α is in rad/sec^2 and t is in seconds. If the angular velocity of the flywheel is 60 rad/sec at the end of 4 seconds, determine the angular velocity at the end of 6 secondds. How many revolutions take place in these 6 seconds?

==== END ====

2×180 $\pi = 180$ 2π 60 rad 360